

TWITTER ANALYSIS – USING POWERBI

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1. Introduction

This report summarizes the development and execution of nine analytical tasks using Power BI, focusing on Twitter data. The project aimed to apply specific business rules, DAX calculations, and dynamic time-based visualizations to build interactive, conditionally-filtered dashboards.

2. Background

The project involved creating advanced Power BI visuals to analyze tweet engagement, impressions, and interaction types. Each task required strict filtering based on tweet metrics (likes, retweets, replies, impressions, word/character count, posting time) and dashboard visibility control depending on the system time (IST).

The goal was to demonstrate expertise in dynamic filtering, DAX measure design, data transformation, and business rule implementation.

3. Learning Objectives

- Understand advanced DAX to filter and conditionally display data.
 - Apply dynamic time-based visibility for visuals (using IST).
 - Handle multiple data validation conditions (even/odd checks, character limits, specific letter exclusions).
 - Manage blank data scenarios professionally in Power BI.
 - Create clean, user-friendly interactive dashboards.
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4. Activities and Tasks

The nine tasks completed were:

Task Description

- 1 Display Top 10% tweets by engagement, filtered by likes >50, weekday post, and tweet character count <30.
- 2 Visualize Average Engagement Rate and Total Impressions for tweets posted between Jan–Jun 2020 with ≥ 100 impressions and 0 likes.
- 3 Identify Top 10 tweets by likes+retweets, posted on weekdays, with even impressions, odd tweet date, word count <30.

Task Description

- 4 Clustered bar chart showing sum of URL, Profile, and Hashtag clicks by tweet category, word count >40.
 - 5 Scatter chart analyzing media engagements vs media views for tweets with >10 replies, engagement >5%, odd tweet dates, word count >50.
 - 6 Line chart showing monthly trend of average engagement rate, comparing tweets with vs without media.
 - 7 Dual-axis chart showing media views vs media engagements by day of the week for last quarter.
 - 8 Compare engagement rate for tweets with vs without app opens, filtered by weekday posts between 9AM–5PM.
 - 9 Clustered bar chart comparing replies, retweets, and likes for tweets with media engagements above median between Jun–Aug 2020.
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5. Skills and Competencies

- Advanced DAX formula writing (PERCENTILEX.INC, SWITCH, IF, CALCULATE).
 - Dynamic time-based chart visibility using NOW() and HOUR().
 - Data cleansing (handling character count, word count, and exclusions).
 - Visual customization (combo charts, scatter, line, clustered column).
 - Handling blank datasets using fallback text messaging techniques.
 - Cross-validation of data (even/odd numbers, letter exclusion).
 - Error troubleshooting and debugging of DAX errors.
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6. Feedback and Evidence

- Successfully implemented dynamic dashboard behavior based on system time (IST).
- Clean handling of blank charts where no data matched strict conditions.
- Effective application of complex business rules to filter the dataset.
- Feedback included identifying situations where conditions were too strict (e.g., no tweets without 'S' or with word count above 50) and creating fallback solutions.

- Dashboards maintained clean and user-friendly appearance even when visuals were empty.

7. Challenges and Solutions

Challenge	Solution
DAX errors using measures in filter pane	Shifted to dynamic visibility control using conditional formatting instead.
No data matching strict filter conditions (e.g., character count >40)	Implemented fallback message cards dynamically using DAX.
Date parsing and quarter filtering errors	Corrected parsing logic, ensured timezone consistency (UTC/IST).
Controlling chart visibility for charts (not just cards)	Used BLANK() strategy inside DAX measures instead of trying to hide visuals.
Heavy data conditions causing blank visuals	Communicated properly to users with friendly text explanations.

8. Outcomes and Impact

- Created fully functional, business-rule-driven Power BI dashboards.
- Mastered advanced DAX and time-based visibility controls.
- Improved handling of special cases when no data matched the conditions.
- Gained confidence to handle dynamic filters, complex visuals, and real-world dashboard requirements.
- Developed clean, dynamic dashboards ready for real-time business reporting under strict data conditions.

9. Conclusion

This Power BI project successfully demonstrated the application of complex data filtering, dynamic time-based visual control, and advanced DAX techniques to create professional dashboards.

Despite challenging data conditions and strict requirements, effective solutions were implemented to ensure the dashboards were interactive, clean, and business-ready.

The project enhanced technical skills significantly and established a strong foundation for solving complex Power BI reporting needs in the future.